DEM 5283/7283: Statistics for Demographic Data II

Course Syllabus, Spring 2027

Corey S. Sparks, PhD

Course information

Course: DEM 5283/7283, Statistics for Demographic Data II
Credit hours: 3
Modality: Traditional in-person
Meeting time: Thursdays, 6:00–8:45 PM
Location: [Room TBD]
Dates: January 21, 2027 to May 6, 2027

Prerequisite: Statistics for Demographic Data I, or an equivalent graduate course in linear regression, or consent of the instructor.

Course description

This course is an in-depth treatment of the generalized linear model framework as demographers use it, including models for binary, ordinal, multinomial and count outcomes, models for longitudinal and multilevel data, and nonlinear regression with splines and GAMs. Model fit, model comparison and regression diagnostics are covered for each method. Students are also introduced to the analysis of complex survey data, decomposition methods, multiple imputation, the foundations of causal inference, tree-based machine learning, and a reproducible research workflow. Every method is illustrated with demographic survey data and applied examples from real research.

Learning objectives

By the end of the course, you will be able to:

1. **Analyze complex survey data correctly.** Specify a survey design with weights, strata and clusters (or replicate weights), and produce estimates and standard errors that describe the population rather than the sample.
2. **Choose and fit the right model for the outcome.** Fit and interpret models for binary, ordinal, multinomial, count, longitudinal and multilevel data, and explain the results on the scale your audience cares about, using predicted probabilities and average marginal effects.
3. **Compare, decompose and diagnose models.** Compare nested models correctly, decompose differences between groups, assess model fit, and handle missing data with multiple imputation.
4. **Reason carefully about causes and predictions.** Draw and use a causal diagram, estimate effects with propensity score weighting, assess sensitivity to unmeasured confounding, and tell when a prediction model is and is not the right tool.
5. **Work reproducibly.** Carry out a complete analysis in R with Quarto, renv and Git, so that someone else can run your files and get the same numbers.

Instructor

Instructor: Corey S. Sparks, PhD
Department: Demography
Email: corey.sparks@utsa.edu
Office: MB 2.248J
Office hours: Thursdays, 3:00–5:00 PM, or by appointment (email me to set one up)

Email is the best way to reach me. I'll usually answer within one business day. Put "DEM 7283" in the subject line so your message doesn't get lost among the rest.

Software

We use R for everything in this course, through Positron. Please install the current release of R (4.6.2 as of this writing), Positron and Git before the first class. On Windows, also install Rtools, which you'll need for week 12.

To install the packages we'll use this semester, paste this line into the Positron console:

```
source("https://raw.githubusercontent.com/coreysparks/Rcode/master/install_first_7283.R")
```

The script skips anything you already have, so it's safe to run again if it gets interrupted. All the course materials, data and exercises are in the course repository at https://github.com/coreysparks/DEM7283. We'll go over the setup, including `renv` and Git, in the first class.

Course materials

There is no single textbook that covers everything we do, so readings come from several books and journal articles. The weekly readings are listed in the schedule below, and the full citations are at the end of this syllabus.

Title	Author	ISBN	Status
Statistical Methods for Categorical Data Analysis, 2nd ed.	Powers and Xie	9780123725622	Required
Exploring Complex Survey Data Analysis Using R	Zimmer, Powell and Velásquez	9781032302867	Recommended
Regression and Other Stories	Gelman, Hill and Vehtari	9781107676510	Recommended
Extending the Linear Model with R, 2nd ed.	Faraway	9781498720960	Recommended
Tidy Modeling with R	Kuhn and Silge	9781492096481	Recommended
Multilevel Modeling	Luke	9781412985147	Recommended
Missing Data	Allison	9780761916727	Recommended

Several excellent books are free online, and I'll point you to chapters from them:

- *R for Data Science*, 2nd ed. (Wickham et al. 2023): https://r4ds.hadley.nz
- *Flexible Imputation of Missing Data*, 2nd ed. (Buuren 2018): https://stefvanbuuren.name/fimd
- *An Introduction to Statistical Learning* (James et al. 2021): https://www.statlearning.com
- *Causal Inference: What If* (Hernán and Robins 2020): https://miguelhernan.org/whatifbook

Journal articles are available through the UTSA Libraries.

Assessments and grading

Component	Weight
Homework (weekly exercises)	60%
Final paper proposal	10%
Final paper presentation	10%
Final paper	20%
Total	**100%**

Homework (60%). Each week's materials end with a set of exercises, and those exercises are the homework. There are 14 sets (weeks 1–7 and 9–15). I drop your lowest score, and the remaining 13 count equally. Each homework is a Quarto document that renders from top to bottom without errors. Submit the rendered HTML file, published with Posit Publisher, and a link to the GitHub repository that holds your homework, through Canvas. Homework is due before class one week after it's assigned. I grade on whether the code runs and is correct, and just as much on whether your written interpretation is correct and says what the numbers mean.

Final paper (40% total). The final paper is an original analysis of demographic data using one or more of the methods from this course. It should read like a short research article: a research question and its motivation, a brief literature review, data and methods, results, and a discussion. I expect about 15–20 double-spaced pages, plus tables and figures, along with the Quarto document and code that produce every number in the paper. The paper has three graded parts:

- **Proposal (10%), due February 25 (week 6).** One to two pages: your research question, the data you'll use, the outcome and key predictors, and the model you expect to fit. Talk to me before then if you're unsure about any of it.
- **Presentation (10%), May 6 (week 16).** A 15-minute presentation of your analysis and results, followed by questions from the class.
- **Paper (20%), due May 6 (week 16).** The written paper and the files that reproduce it.

Grade distribution

Letter grade	Grade range
A	90–100
B	80–89
C	70–79
D	60–69
F	0–59

Course schedule

Readings should be done before the class meeting in the week they're listed. The materials for each week are in the `weeks/` folder of the course repository.

Week	Date	Topic	Due
1	Jan 21	Course introduction; reproducible workflow with Quarto, renv and Git	

Week	Date	Topic	Due
2	Jan 28	Complex survey design, weights, replicate variance	Homework 1
3	Feb 4	Binary outcomes: logit and probit, predicted probabilities, average marginal effects	Homework 2
4	Feb 11	Comparing nested logit models (KHB); prediction and cross validation	Homework 3
5	Feb 18	Ordinal and multinomial models	Homework 4
6	Feb 25	Counts and rates: Poisson, negative binomial, offsets, zero inflation	Homework 5; **paper proposal**
7	Mar 4	Decomposition analysis: Kitagawa and Oaxaca-Blinder	Homework 6
8	Mar 11	**Spring break, no class**	
9	Mar 18	Missing data and multiple imputation with survey designs	Homework 7
10	Mar 25	Longitudinal models: GEE, fixed effects, within-between models	Homework 9
11	Apr 1	Nonlinear regression: splines and GAMs	Homework 10
12	Apr 8	Multilevel models I: centering, random slopes, level-specific weights	Homework 11
13	Apr 15	Multilevel models II: GLMMs, growth curves, age-period-cohort models	Homework 12
14	Apr 22	Causal inference foundations: DAGs and propensity score weighting	Homework 13
15	Apr 29	Tree-based machine learning with tidymodels	Homework 14
16	May 6	**Final paper presentations**	Homework 15; **final paper**

Homework is numbered by the week it's assigned, so there is no Homework 8.

Weekly readings

Week 1. Reproducible workflow. Wickham et al. (2023), the chapters on workflow and projects; Wickham et al. (2019).

Week 2. Complex survey design. Zimmer et al. (2024), chapters 2 and 10; Kish (1992); Centers for Disease Control and Prevention (2025).

Week 3. Logit and probit. Mood (2010); Breen et al. (2018).

Week 4. Nested logits and prediction. Allison (1999); Karlson et al. (2012); Lumley and Scott (2015); Kuhn and Silge (2022), the chapters on resampling.

Week 5. Ordinal and multinomial models. Long and Freese (2014), the chapters on ordinal and nominal outcomes.

Week 6. Counts and rates. Zeileis et al. (2008); Cameron and Trivedi (2013), chapters 1–3; Lambert (1992).

Week 7. Decomposition. Kitagawa (1955); Oaxaca (1973); Jann (2008); Fairlie (2005).

Week 9. Missing data. Buuren (2018), chapters 1–2; von Hippel (2020); Reiter et al. (2006).

Week 10. Longitudinal models. Allison (2009); Bell and Jones (2015); Liang and Zeger (1986).

Week 11. Splines and GAMs. Wood (2017), chapters 4–5; Harrell (2015), chapter 2.

Week 12. Multilevel models I. Enders and Tofighi (2007); Rabe-Hesketh and Skrondal (2006).

Week 13. Multilevel models II. Austin and Merlo (2017); Singer and Willett (2003), chapters 3–4; Fosse and Winship (2019); Bell and Jones (2014).

Week 14. Causal inference. Hernán and Robins (2020), chapters 1–7; Austin (2011); Elwert and Winship (2014); VanderWeele and Ding (2017).

Week 15. Machine learning. James et al. (2021), chapter 8; Kuhn and Silge (2022), chapters 5–7; Molina and Garip (2019); Obermeyer et al. (2019).

Course policies

Attendance and participation

I expect you to come to every class. Most of each meeting is working through the week's analysis together, and that's hard to make up from the slides. If you have to miss a class, let me know beforehand when you can, and then work through that week's materials on your own and come see me with questions.

Questions in class are always welcome, and the ones you think are basic are usually the ones half the room was wondering about. I expect everyone to treat classmates with respect, in disagreements about methods and in everything else.

Late work

Homework turned in up to one week late loses 10%. After that I won't accept it unless we've made an arrangement in advance, and the dropped homework is there for the week when something goes wrong. The final paper and presentation can't be late, because they fall at the end of the semester. If you have a serious problem (illness, a family emergency), talk to me as early as you can, and we'll work something out.

Collaboration

You're welcome to work with classmates on the homework, and I encourage it. But each of you writes and submits your own code and your own answers. If you worked with someone, put their names at the top of your homework. The final paper is individual work.

Guidelines for the use of generative AI

You may use generative AI tools to help you write or debug R code. They are genuinely useful for explaining an error message or a block of code, and they also make up function arguments with total confidence, so run what they give you and read the help page for anything you don't recognize.

Everything you write in prose must be your own work, in your own words. That includes the interpretation of results in your homework, the paper proposal, the final paper and your presentation. Never paste restricted or confidential data (for example, data covered by a data use agreement) into an AI tool.

Instructor-initiated drops

Academic freedom[1] is a cornerstone of the University. Academic freedom in its teaching aspect is fundamental for the protection of the rights of the teacher in teaching and of the student to freedom in learning.[2] Each faculty member is entitled to full freedom in the classroom discussing the subject that the faculty member teaches.[3] The University of Texas at San Antonio will not penalize or discipline members of the faculty because of their exercise of academic freedom.

Along with this freedom comes responsibility. It is the responsibility of faculty members to ensure that topics taught are related to the classroom subject. Students should be free to take reasoned exception to the data or views offered in any course of study and to reserve judgment about matters of opinion, but they are responsible for learning the content of any course of study for which they are enrolled.[4] It is not the proper role of the university or any outside agency to attempt to shield individuals from ideas and opinions they find unwelcome, disagreeable, or even deeply offensive.[5] Engaging with new ideas and perspectives helps students grow intellectually and is beneficial to the educational process.

Our commitment to inclusivity

The University of Texas at San Antonio, a Hispanic Serving Institution situated in a global city that has been a crossroads of peoples and cultures for centuries, values diversity and inclusion in all aspects of university life. As an institution expressly founded to advance the education of Mexican Americans and other underserved communities, our university is committed to promoting access for all. UTSA, a premier public research university, fosters academic excellence through a community of dialogue, discovery and innovation that embraces the uniqueness of each voice.

Syllabus changes

The syllabus is subject to change at the instructor's discretion. Any changes or corrections to the course materials, assignment dates or other policies will be announced in class and on Canvas ahead of time. You are responsible for checking Canvas for updates to the syllabus.

Essential student information

- **Important:** Bookmark and visit the Common Syllabus Information page (https://provost.utsa.edu/syllabus.html). It has information on counseling services, minor medical issues, supplemental instruction, tutoring, academic success coaching, sexual harassment and sexual misconduct, campus safety and emergency preparedness, and the Roadrunner Creed.
- **Disability accommodations:** UTSA provides reasonable accommodations to students through Student Disability Services. For eligibility, policies and requirements, visit https://www.utsa.edu/disability or call (210) 458-4157.

[1] Statement adapted from Texas A&M University's Syllabus Statement Regarding Academic Freedom.
[2] 1940 Statement of Principles on Academic Freedom and Tenure.
[3] Board of Regents Rule 31004; HOP Policy 4.02.
[4] American Association of University Professors, Joint Statement on Rights and Freedoms of Students.
[5] The Chicago Statement.

- **Academic integrity:** You are responsible for making sure your work is consistent with UTSA's standards for academic integrity. Review Section 203 of the UTSA Student Code of Conduct for those standards.
- **Student services:** UTSA provides many services for students, from counseling to tutoring to a food pantry. See Student Affairs Programs and Services and Student Success for more information.
- **Libraries:** The UTSA Libraries have journals, research tutorials and technology you can borrow, and a librarian for your department.
- **Technical support:** For technical requirements, support and resources, see Academic Innovation's Student Technical Support page.

References

Allison, Paul D. 1999. "Comparing Logit and Probit Coefficients Across Groups." *Sociological Methods & Research* 28 (2): 186–208. https://doi.org/10.1177/0049124199028002003.

Allison, Paul D. 2009. *Fixed Effects Regression Models*. Quantitative Applications in the Social Sciences 160. Sage.

Austin, Peter C. 2011. "An Introduction to Propensity Score Methods for Reducing the Effects of Confounding in Observational Studies." *Multivariate Behavioral Research* 46 (3): 399–424.

Austin, Peter C., and Juan Merlo. 2017. "Intermediate and Advanced Topics in Multilevel Logistic Regression Analysis." *Statistics in Medicine* 36 (20): 3257–77.

Bell, Andrew, and Kelvyn Jones. 2014. "Another 'Futile Quest'? A Simulation Study of Yang and Land's Hierarchical Age-Period-Cohort Model." *Demographic Research* 30 (11): 333–60.

Bell, Andrew, and Kelvyn Jones. 2015. "Explaining Fixed Effects: Random Effects Modeling of Time-Series Cross-Sectional and Panel Data." *Political Science Research and Methods* 3 (1): 133–53.

Breen, Richard, Kristian Bernt Karlson, and Anders Holm. 2018. "Interpreting and Understanding Logits, Probits, and Other Nonlinear Probability Models." *Annual Review of Sociology* 44: 39–54.

Buuren, Stef van. 2018. *Flexible Imputation of Missing Data*. 2nd ed. CRC Press. https://stefvanbuuren.name/fimd/.

Cameron, A. Colin, and Pravin K. Trivedi. 2013. *Regression Analysis of Count Data*. 2nd ed. Cambridge University Press.

Centers for Disease Control and Prevention. 2025. *2024 SMART BRFSS MMSA Methodology*. https://www.cdc.gov/brfss/annual_data/2024/pdf/2024_SMART_BRFSS_MMSA_Methodology-508.pdf.

Elwert, Felix, and Christopher Winship. 2014. "Endogenous Selection Bias: The Problem of Conditioning on a Collider Variable." *Annual Review of Sociology* 40: 31–53.

Enders, Craig K., and Davood Tofighi. 2007. "Centering Predictor Variables in Cross-Sectional Multilevel Models: A New Look at an Old Issue." *Psychological Methods* 12 (2): 121–38.

Fairlie, Robert W. 2005. "An Extension of the Blinder-Oaxaca Decomposition Technique to Logit and Probit Models." *Journal of Economic and Social Measurement* 30 (4): 305–16.

Fosse, Ethan, and Christopher Winship. 2019. "Analyzing Age-Period-Cohort Data: A Review and Critique." *Annual Review of Sociology* 45: 467–92.

Harrell, Jr., Frank E. 2015. *Regression Modeling Strategies*. 2nd ed. Springer.

Hernán, Miguel A., and James M. Robins. 2020. *Causal Inference: What If*. Chapman & Hall/CRC.

James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. 2021. *An Introduction to Statistical Learning with Applications in R*. 2nd ed. Springer.

Jann, Ben. 2008. "The Blinder-Oaxaca Decomposition for Linear Regression Models." *The Stata Journal* 8 (4): 453–79.

Karlson, Kristian Bernt, Anders Holm, and Richard Breen. 2012. "Comparing Regression Coefficients Between Same-Sample Nested Models Using Logit and Probit: A New Method." *Sociological Methodology* 42 (1): 286–313. https://doi.org/10.1177/0081175012444861.

Kish, Leslie. 1992. "Weighting for Unequal P_i." *Journal of Official Statistics* 8 (2): 183–200.

Kitagawa, Evelyn M. 1955. "Components of a Difference Between Two Rates." *Journal of the American Statistical Association* 50 (272): 1168–94.

Kuhn, Max, and Julia Silge. 2022. *Tidy Modeling with R: A Framework for Modeling in the Tidyverse*. O'Reilly Media. https://www.tmwr.org.

Lambert, Diane. 1992. "Zero-Inflated Poisson Regression, with an Application to Defects in Manufacturing." *Technometrics* 34 (1): 1–14.

Liang, Kung-Yee, and Scott L. Zeger. 1986. "Longitudinal Data Analysis Using Generalized Linear Models." *Biometrika* 73 (1): 13–22.

Long, J. Scott, and Jeremy Freese. 2014. *Regression Models for Categorical Dependent Variables Using Stata*. 3rd ed. Stata Press.

Lumley, Thomas, and Alastair Scott. 2015. "AIC and BIC for Modeling with Complex Survey Data." *Journal of Survey Statistics and Methodology* 3 (1): 1–18.

Molina, Mario, and Filiz Garip. 2019. "Machine Learning for Sociology." *Annual Review of Sociology* 45: 27–45.

Mood, Carina. 2010. "Logistic Regression: Why We Cannot Do What We Think We Can Do, and What We Can Do about It." *European Sociological Review* 26 (1): 67–82. https://doi.org/10.1093/esr/jcp006.

Oaxaca, Ronald. 1973. "Male-Female Wage Differentials in Urban Labor Markets." *International Economic Review* 14 (3): 693–709.

Obermeyer, Ziad, Brian Powers, Christine Vogeli, and Sendhil Mullainathan. 2019. "Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations." *Science* 366 (6464): 447–53.

Rabe-Hesketh, Sophia, and Anders Skrondal. 2006. "Multilevel Modelling of Complex Survey Data." *Journal of the Royal Statistical Society: Series A* 169 (4): 805–27.

Reiter, Jerome P., Trivellore E. Raghunathan, and Satkartar K. Kinney. 2006. "The Importance of Modeling the Sampling Design in Multiple Imputation for Missing Data." *Survey Methodology* 32 (2): 143–49.

Singer, Judith D., and John B. Willett. 2003. *Applied Longitudinal Data Analysis: Modeling Change and Event Occurrence*. Oxford University Press.

VanderWeele, Tyler J., and Peng Ding. 2017. "Sensitivity Analysis in Observational Research: Introducing the E-Value." *Annals of Internal Medicine* 167 (4): 268–74.

von Hippel, Paul T. 2020. "How Many Imputations Do You Need? A Two-Stage Calculation Using a Quadratic Rule." *Sociological Methods & Research* 49 (3): 699–718.

Wickham, Hadley, Mara Averick, Jennifer Bryan, et al. 2019. "Welcome to the tidyverse." *Journal of Open Source Software* 4 (43): 1686.

Wickham, Hadley, Mine Çetinkaya-Rundel, and Garrett Grolemund. 2023. *R for Data Science*. 2nd ed. O'Reilly Media. https://r4ds.hadley.nz.

Wood, Simon N. 2017. *Generalized Additive Models: An Introduction with R*. 2nd ed. CRC Press.

Zeileis, Achim, Christian Kleiber, and Simon Jackman. 2008. "Regression Models for Count Data in R." *Journal of Statistical Software* 27 (8): 1–25.

Zimmer, Stephanie A., Rebecca J. Powell, and Isabella C. Velásquez. 2024. *Exploring Complex Survey Data Analysis Using R: A Tidy Introduction with srvyr and survey*. CRC Press. https://tidy-survey-r.github.io/tidy-survey-book/.